

Emotion Classification Report

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1 Introduction

While traditional sentiment analysis focuses on polarity, this study explores deeper emotional complexity through multi-class classification. Using the GoEmotions dataset, we evaluate four pre-trained language models and implement an ensemble voting strategy. Our approach aims to combine the strengths of diverse architectures to enhance classification performance and provide a more refined analysis of emotional expressions in text.

2 Methodology

2.1 Dataset

In the classification, we use the GoEmotions Dataset [1] developed by Google Research. The 27 original emotions are mapped to 7 Ekman categories.

2.2 Models & Training Configurations

We evaluate four pre-trained language models: DistilBERT [2], RoBERTa [3], DeBERTa [4], and Qwen [5]. The classification head of each model is initialized with an output dimension of $K = 7$, matching the number of classes in the task. Table 1 summarizes their training configurations.

Table 1: Model Training Configurations

Model	Train BS	Eval BS	Weight Decay	Mixed Precision	LoRA
DistilBERT	64	128	0	fp16	None
RoBERTa-large	32	64	0.01	fp16	None
DeBERTa-v3-base	32	128	0.01	None	None
Qwen3.5-4B	16	128	0.01	bf16	True

All models are trained for 4 epochs using the AdamW optimizer with learning rate of $2e-5$. Qwen3.5 is using LoRA with rank $r = 16$, scaling factor $\alpha = 32$, and dropout 0.1, applied to the attention projection modules (q_proj , k_proj , v_proj , o_proj). DistilBERT uses dropout rate of 0.3 rather than weight decay.

Training is conducted on a workstation equipped with an Intel Core i9-13900K CPU and an NVIDIA GeForce RTX 4090 D GPU (48 GB VRAM).

2.3 Evaluation Metrics & Ensemble Voting

The metrics for evaluation are the accuracy score and the macro-f1 score. The classification uses mode as the voting method for ensemble learning.

3 Results

Figure 1, figure 2, figure 3 and figure 4 show the training and evaluation results of each models.

All models achieved a macro F1 score above 0.6 before 1500 training steps, after reaching top validation F1 scores, the scores decrease slowly along with fluctuation. The result shows that all the

pre-trained models can learn the pattern in comments of each emotion easily, but the models are too complicated for the dataset.

The loss figures indicate that RoBERTa, Qwen3 and DeBERTa-v3 converge faster than DistilBERT and achieve much lower losses (under 1.0). For stability, the RoBERTa is more unstable than other models, for it quickly overfitting after epoch 3, the loss has sharp rebound, and the f1 score drops significantly.

Table 2: The Validation Score of Models

Model	Accuracy	Recall	Precision	Macro-F1
DistilBERT	0.7119	0.6610	0.6408	0.6495
RoBERTa-large	0.7046	0.6501	0.6189	0.6321
DeBERTa-v3	0.7208	0.6403	0.6489	0.6434
Qwen3.5-4B	0.7216	0.6370	0.6708	0.6523
Ensemble (Voting)	0.7289	0.6667	0.6703	0.6672

As shown in Table 2, the voting method achieves the highest accuracy of 0.7289 and macro-F1 score of 0.6672, outperforming all individual models. This represents a relative improvement of 2.28% in macro-F1 over the best single model (Qwen3.5-4B), demonstrating the effectiveness of the model ensemble for emotion classification. The voting method enables recall to improve (0.86%), mainly enhancing the ability to detect the true labels. The confusion matrix 5 shows that the ensemble model performs well in most of the labels, while exists misclassification in the ambiguity labels "neutral" with "anger", "joy" and "surprise" labels.

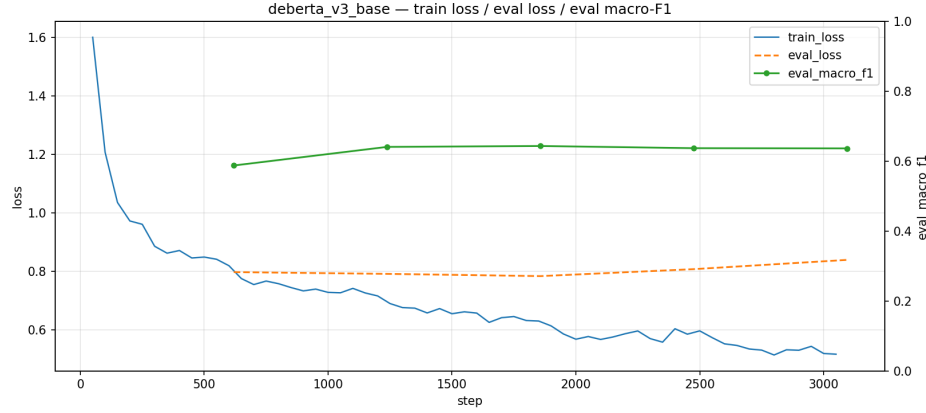


Figure 1: The training loss, evaluation loss, evaluation macro-f1 score of DeBERTa-v3

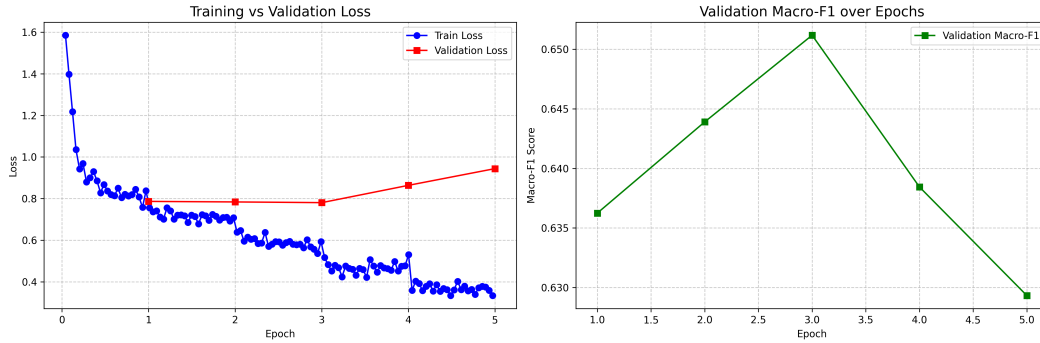


Figure 2: The training and evaluation loss, training and evaluation macro-f1 score of RoBERTa

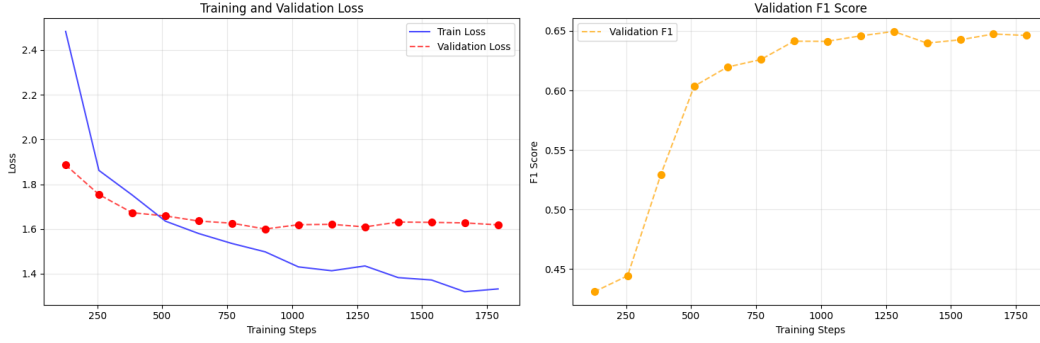


Figure 3: The training and evaluation loss, evaluation macro-f1 score of DistilBERT

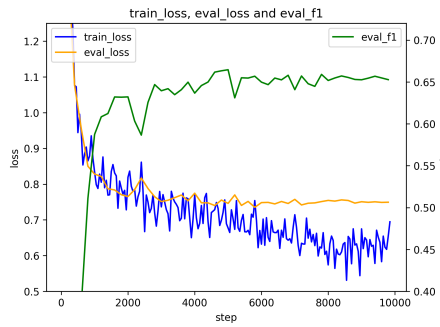


Figure 4: The training and evaluation loss, evaluation macro-f1 score of Qwen3.5-4b

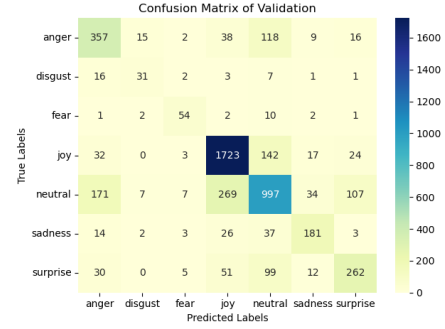


Figure 5: Concrete and Constructions

4 Conclusion

This study compared several pre-trained models (DistilBERT, RoBERTa-large, DeBERTa-v3, and Qwen3.5-4B) on the GoEmotions dataset. Experimental results indicate that while individual models perform competently, they face varying degrees of overfitting and trade-offs between precision and recall.

The ensemble voting mechanism achieved the highest performance, with an accuracy of 0.7289 and a macro-F1 of 0.6672. Notably, the ensemble approach significantly improved recall (0.6667) and reduced Type II errors compared to any single model. This demonstrates that combining diverse architectures effectively enhances the robustness and coverage of emotion classification.

References

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